

Probabilistic Models

Probabilistic model: mathematical description of an uncertain situation
Random experiment: experiment in which the outcome varies in an unpredictable fashion when experiment repeated under the same condition.

↳ specify random experiment by stating experimental procedure and what is measured

↳ examples:

experiment E_1 : select a ball from an urn containing balls numbered from 1 to 50. Note the number of the ball.

experiment E_2 : toss a coin 3 times and note the # of heads.

experiment E_3 : toss a coin 3 times and note the sequence of heads and tails.

Sample space: Set of all possible outcomes of an experiment denote by Ω

$$S_1 = \{1, 2, \dots, 50\}$$

$$S_2 = \{0, 1, 2, 3\}$$

$$S_3 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Event: collection of possible outcomes.

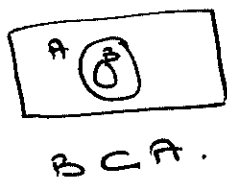
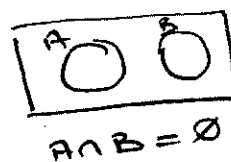
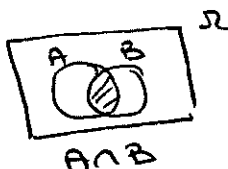
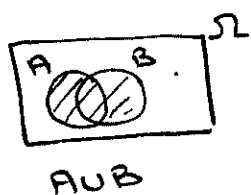
event E_{11} : "an even-numbered ball is selected"
 $E_{11} = \{2, 4, 6, \dots, 48, 50\}$

event E_{21} : "# of heads = # of tails"
 $E_{21} = \emptyset$

event E_{31} : "three tosses give the same outcome"
 $E_{31} = \{HHH, TTT\}$

Set operations: combine events using set operations to obtain other events.

(2)



1) Commutative properties:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

2) Associative properties:

$$A \cup (B \cap C) = (A \cup B) \cap C, \quad A \cap (B \cup C) = (A \cap B) \cup C$$

3) Distributive properties:

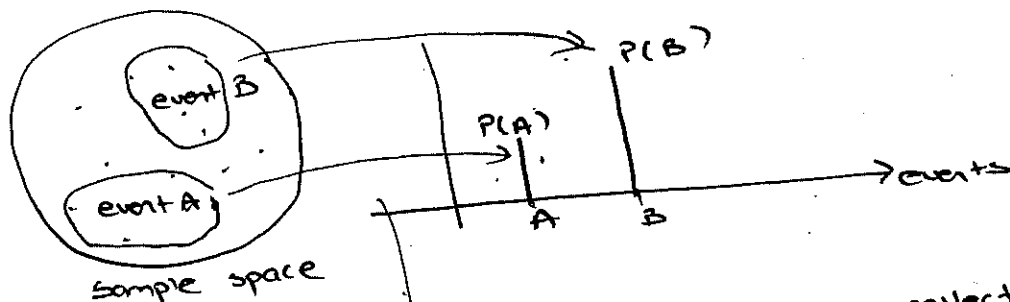
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4) De Morgan's rules:

$$(A \cap B)^c = A^c \cup B^c, \quad (A \cup B)^c = A^c \cap B^c$$

Probability Law:

random experiment



probability law: encodes the collective likelihood of elements in A .

Probability Axioms:

1) $P(A) \geq 0$ for any event A .

2) A and B disjoint events ($A \cap B = \emptyset$) $\Rightarrow P(A \cup B) = P(A) + P(B)$

more generally: $A_1, A_2, \dots$ disjoint events $\Rightarrow$

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

3) $P(\Omega) = 1$

Properties of Probability Laws:

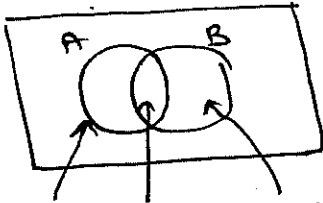
$$1. P(A^c) = 1 - P(A).$$

Proof: $A \cap A^c = \emptyset$

axiom 2: $P(A \cup A^c) = P(A) + P(A^c)$

$$\Omega = A \cup A^c \Rightarrow P(A \cup A^c) = P(\Omega) = 1 = P(A) + P(A^c)$$

$$2. P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



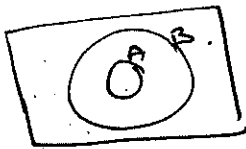
$A \cap B^c$ $A \cap B$ $A^c \cap B$ ← disjoint events

$$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B)$$

$$P(A) = P(A \cap B^c) + P(A \cap B)$$

$$P(B) = P(A \cap B) + P(A^c \cap B) \quad \Rightarrow \quad P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

$$3. A \subset B \Rightarrow P(A) \leq P(B)$$



$$B = A \cup (A^c \cap B)$$

$$P(B) = P(A) + P(A^c \cap B) \geq P(A) \geq 0$$

Discrete Sample Spaces

Suppose sample space consists of a finite # of possible outcomes

$$\Omega = \{a_1, a_2, \dots, a_n\} \leftarrow \text{all elements are mutually exclusive}$$

Probability of any event $B = \{a_1', a_2', \dots, a_m'\}$

$$P(B) = P(\{a_1', a_2', \dots, a_m'\})$$

$$= P(\{a_1'\}) + P(\{a_2'\}) + \dots + P(\{a_m'\})$$

If $P(a_1) = P(a_2) = \dots = P(a_n) = \frac{1}{n}$

$$P(B) = \frac{m}{n}$$

← # of outcomes in the event
← total # of outcomes in sample space.

x: Consider the experiment of rolling a pair of 4-sided ^④ dice.

$$P(\{\text{sum of the rolls is even}\}) = ?$$

Sample space: $\{(i, j) \mid \begin{array}{l} i: \text{result of first roll,} \\ j: \text{result of second roll} \end{array}\}$.

$$= \{(1,1), (1,2), (1,3), (1,4), \\ (2,1), (2,2), (2,3), (2,4), \\ (3,1), (3,2), (3,3), (3,4), \\ (4,1), (4,2), (4,3), (4,4)\}$$

event: $\{(i, j) \mid \begin{array}{l} i: \text{result of first roll,} \\ j: \text{result of second roll,} \\ \text{sum of the roll is even} \end{array}\}$

$$= \{(1,1), (1,3), \\ (2,2), (2,4), \\ (3,1), (3,3), \\ (4,2), (4,4)\}.$$

$$P(\{\text{sum of rolls even}\}) = \frac{8}{16} = \frac{1}{2}.$$

Continuous Sample Spaces:

events of interests: intervals of real line or rectangular regions.

ex: pick a number x at random between zero and one

$$S = [0, 1].$$

If all outcomes s are equally likely to be selected,

$$P([0, \frac{1}{2}]) = \frac{1}{2}.$$

$$P([a, b]) = b - a, \quad 0 \leq a \leq b \leq 1.$$

ex: pick two numbers x and y at random between 0 and 1



Assume all pairs of numbers in unit square are equally likely.